



NATIONAL SCHOOL OF BUSINESS MANAGEMENT

BSc (Hons) in Software Engineering – 17.2/18.1

BSc (Hons) in Computer Science – 17.2/18.1

4th Year 2nd Semester Examination

24th October 2021

CS404.3 - Parallel and Distributed Computing

Instructions to Candidates

- 1) **All questions in section A (40 Marks) AND Any three questions from section B (60 Marks)**
- 2) **The duration of the question paper is 5 hours. Including half an hour to download the paper and upload your answers in a single file. (Note: No email submissions are accepted under any condition.)**
- 3) Weightage of Examination: 60% out of final grade
- 4) Download the paper, provide answers to the selected questions in a word document.
- 5) Please upload the document with answers (Answer Script) to the submission link before the submission link expires
- 6) Answer script should be uploaded in PDF Format
- 7) Under any circumstances E-mail submissions would not be taken into consideration for marking. Incomplete attempt would be counted as a MISSED ATTEMPT.
- 8) The Naming convention of the answer script – Module Code_Subject name_Index No
- 9) You must adhere to the online examination guidelines when submitting the answer script to N-Learn.
- 10) Your answers will be subjected to Turnitin similarity check, hence, direct copying and pasting from internet sources, friend's answers etc. will be penalized.

Section A: Answer all questions

Q1)

(a) Explain the terms in relation to SOA

a. UDDI

b. WSDL

(4 marks)

(b) Compare SOAP and RESTful web services

(6 marks)

Q2)

Explain the purpose of different layers in RMI for invoking a method of a remote object. Supply a suitable architectural diagram in your answer.

(10 marks)

Q3)

a. What is parallel computing

(4 Marks)

b. Explain 4 possible classifications Flynn's Classical Taxonomy model

(6 Marks)

Q4)

a. Explain two differences between multi-core and the multiprocessing

(4 Marks)

b. What's the difference between SMP and MPP?

(6 Marks)

Section B: Answer any THREE questions

Q5)

(a) Describe two essential characteristics of Distributed Systems

(04 Marks)

(b) With the aid of a diagram, describe the main components of the architecture of a distributed system

(08 Marks)

(c) Middleware delivers an abstract layer of support that acts as a building block for the construction of the target application(s). What is meant by this statement?

(04 Marks)

(d) Explain the terms Availability and Redundancy.

(04 Marks)

Q6)

(a) Answer the following in relation to the Google file system

- a. Describe the purpose of the operation log which is maintained by master server of the GFS
(03 Marks)
- b. What is the purpose of the chunk server
(03 Marks)
- c. Describe the principle of Autonomic Computing involved in the Google File System
(04 Marks)

b) In relation to distributed transaction

- a. Explain the term atomic
(03 marks)
- b. using a proper diagram explain how a distributed transaction based on two phase commit protocol mechanism is faithful to the atomicity principle
(07 Marks)

Q7)

- a) Enterprise Application Integration is the process of bringing data or a function from one application program together with that of another application program. Explain three reasons why enterprise integration is essential
(06 Marks)
- b) Using a practical example explain how to use the file transfer approach in enterprise integration. Also discuss two issues concerning the files that must be addressed for this approach to work.
(04 Marks)

- c) Describe two advantages of virtualization for a business IT center.
(04 Marks)
- d) Describe the responsibilities of each tier in 3-tier architecture
(6 marks)

Q8)

- a. What is GPU computing?
(4 marks)
- b. Compare GPU and CPU computing
(6 marks)
- c. "GPUs are commonly used for non-graphical tasks such as machine learning and scientific computation"
Do you agree with this statement? Explain your reasons
(6 marks)
- d. How does parallel programming differ from multithreaded programming?
(4 marks)

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